Pattern Recognition Mock Exam

Full Name	
Matriculation Number	
Course of Studies	
I have read all the information below and entered required data truthfully:	
Signature	

- You have 90 minutes to finish the exam.
- All questions are single or multiple choice. For each multiple-choice question, one or more than one option could be correct.
- This exam does not include negative points. The minimum points you can get from a question is zero. You need to choose all correct options to get full grade points. Choosing any incorrect option will be penalized.
- Only the answers marked on the answer sheet will be evaluated. You can use this question sheet's space to write down any calculation or note you want for your own analysis. Nothing that you write in this question sheet will be considered. The final grade is evaluated solely based on your answer sheet.
- You are **not allowed** to answer with a pencil.
- You are not allowed to use any electronic auxiliaries including calculators.
- You are allowed to use exactly one **handwritten** DinA4 sheet of notes. (Back and front, photocopies are not allowed.)

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PLEASE WRITE ANSWERS ON THE ANSWER SHEET!

Section	1	2	3	4	5	6	7	Sum
Points	6	8	8	8	8	6	8	52

Answersheet Guideline

1. Mark your answers clear in the designated spaces.
2. Only **pens** are allowed for filling in answers in answer sheets. Do not use pencils, highlighters, or correction fluid.
3. The correct method for marking answers is shown in Figure 1. Only use a **cross-mark** on the **answer sheet** to indicate your choice. If you cannot correct your answer further in the exam, request a new answer sheet.
4. The exam will be graded automatically. Ensure that your markings are clear and confined within the designated spaces.

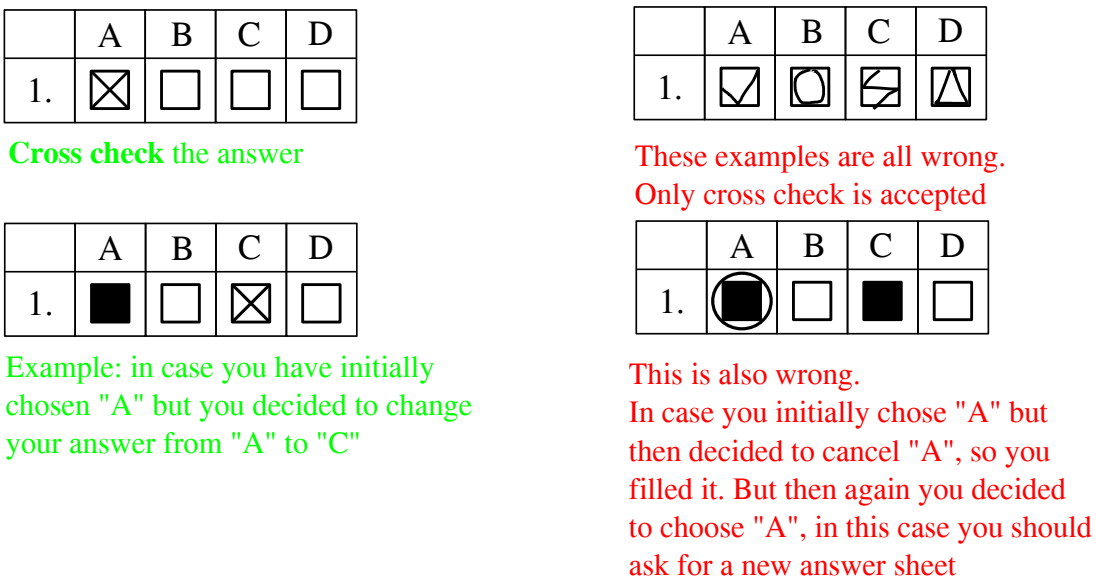

Figure 1: Instructions on the correct and wrong behavior of cross-check in answer sheet.

1 Introduction

Question 1.1 1 P.

We discussed the 6 fundamental postulates of pattern recognition. Postulate 6 states "Similarity through feature space proximity". Which of the following statements best describes this postulate? (SC)

- ☐ A. Two patterns are similar if their features differ only lightly. This implies that small differences in the input features should not change the classification drastically.
- ☐ B. Patterns belonging to the same class form a compact group in the feature space, meaning their features are similar to each other.
- ☐ C. A pattern consists of simpler constituents which have certain relations to each other and can be decomposed into these constituents.
- ☐ D. A complex pattern has a certain structure, meaning that not any arrangement of simpler constituents is a valid pattern.

Solution 1.1

A
B describes Postulate 3 ("Compactness and separability").
C describes Postulate 4 ("Decomposability").
D describes Postulate 5 ("Structural constraints")

Question 1.2 1 P.

Why is convexity an important property in optimization problems? (SC)

- ☐ A. It guarantees that the problem has a closed-form analytical solution, eliminating the need for iterative algorithms like Gradient Descent.
- ☐ B. It ensures that any local minimum is also a global minimum, preventing the optimization algorithm from getting stuck in local optima.
- ☐ C. It implies that the objective function is linear, which ensures that the computational complexity of the optimization is always linear.
- ☐ D. It ensures that the Hessian matrix is always indefinite, allowing the algorithm to escape saddle points more efficiently.

Solution 1.2

B

Question 1.3 4 P.

State which of the 4 following statements are true for discriminative and/or generative models: (MC)

	1. Discriminative	2. Generative
A. It directly models the decision boundary:	☐	☐
B. It can classify an input feature vector:	☐	☐
C. It models class distributions:	☐	☐
D. It can be trained with training data:	☐	☐

Solution 1.3

1. Discriminative model
2. Discriminative and Generative model
3. Generative model
4. Discriminative and Generative model

2 Logistic Regression

You are a data analyst for a municipal health agency. To better allocate medical resources, you are building a model to predict the probability that a resident will contract a seasonal respiratory illness ($y = 1$) based on factors like age, air quality index, and vaccination status (x). You have chosen **Logistic Regression** because of its ability to provide interpretable, calibrated probabilities. (8 P.))

Question 2.1 1 P.

As you explain the model choice to the Health Commissioner, which of the following statements correctly describe the fundamental properties of your Logistic Regression model?

☐ A. It is a generative model that estimates the joint density $p(x, y)$ of the population's health data before applying Bayes' rule.

☐ B. It is a discriminative model that directly estimates the risk $p(y|x)$ without modeling how environmental features $p(x|y)$ are distributed.

☐ C. Unlike a simple Perceptron, it outputs a calibrated probability value between 0 and 1, allowing for nuanced risk tiers.

☐ D. It requires the assumption that the feature distributions for healthy and ill residents are Gaussian with equal covariance matrices.

☐ E. It is solved using a closed-form analytical solution, making training instantaneous once the data is collected.

Solution 2.1

B, C

Question 2.2 1 P.

To map health scores to probabilities, you use the logistic sigmoid function $g(z) = \frac{1}{1+e^{-z}}$. Which of the following mathematical properties of this "risk curve" are valid?

☐ A. The function satisfies the symmetry relation $g(-z) = -g(z)$.

☐ B. The sensitivity (derivative) can be expressed using the risk value itself: $g'(z) = g(z)(1 - g(z))$.

☐ C. The function maps any real-valued health score strictly monotonically to a probability in $(0, 1)$.

☐ D. At a neutral score of 0, the predicted risk is $g(0) = 0$.

☐ E. The function satisfies the symmetry relation $g(-z) = 1 - g(z)$.

Solution 2.2

B, C, E

Question 2.3 2 P.

You define a "High Risk" alert when $p(y = 1|x) \geq 0.5$. Which statements regarding the geometric boundary between "At Risk" and "Safe" zones are correct?

☐ A. The decision boundary is the set of points where the log-odds (logit) of falling ill equal 0.

☐ B. The decision boundary is defined by the hyperplane $\theta^T x = 0.5$.

☐ C. The decision boundary is linear in the feature space defined by x.

☐ D. If we use non-linear transformations $\phi(x)$ (e.g., squaring the age), the decision boundary must still be linear in the original input space.

☐ E. Residents lying exactly on the decision boundary have a predicted risk probability of 0.5.

Solution 2.3

A, C, E

Question 2.4 2 P.

You train the model by maximizing the log-likelihood $L(\theta)$ of the historical health records. Which statements about the optimization and training process are true?

☐ A. The log-likelihood function $L(\theta)$ is concave, ensuring that your gradient ascent will find the global best fit for the data.

☐ B. The gradient of the log-likelihood is $\nabla L(\theta) = \sum_{i=1}^{m}(y_i - g(\theta^T x_i))x_i$.

☐ C. The model is prone to getting stuck in local optima if the initial health weights are poorly chosen.

☐ D. The Newton-Raphson update step used for training can be interpreted as an Iteratively Reweighted Least Squares (IRLS) problem.

☐ E. The weight update depends only on whether the prediction was "right or wrong" (the sign), similar to the Perceptron algorithm.

Solution 2.4

A, B, D

Question 2.5 2 P.

Consider a simplified model with weights for Age (x_1), Exposure (x_2), and a Bias: $\theta = [1, 1, -2]^T$. The risk score is $z = 1 \cdot x_1 + 1 \cdot x_2 - 2$. Which of the following are true?

☐ A. A resident with Age=1 and Exposure=1 lies exactly on the decision boundary.

☐ B. A resident with Age=2 and Exposure=2 is assigned a risk probability $p(y = 1|x) > 0.5$.

☐ C. A resident with Age=0 and Exposure=0 has a log-odd risk score of 0.

☐ D. The equation of the risk decision boundary is $x_2 = -x_1 + 2$.

☐ E. If the bias term were changed to 0, a "newborn" (Age=0) with zero exposure would lie on the decision boundary.

Solution 2.5

A, B, D, E

3 Evaluation

You are lead systems engineer for an airport security project. Your team is testing a new AI-based x-ray scanner designed to detect prohibited items (Positive class) in passenger luggage. To ensure the safety of the airport while minimizing delays for passengers, you must rigorously evaluate your classifier using confusion matrices, ROC curves, and multi-class metrics for different types of prohibited items. (8 P.)

Question 3.1 1 P.

A prototype scanner is evaluated on a test set of $N = 100$ bags. The following confusion matrix summarizes the initial results:

reference\hypothesis	+	−	
+	40	10	50
−	15	35	50
	55	45	100

Which statements are correct based on this audit?

☐ A. $\text{FPR} = \frac{15}{55}$

☐ B. $\text{ACC} = \frac{75}{100}$

☐ C. TPR equals Recall and here $\text{TPR} = \frac{40}{50}$

☐ D. The negative predictive value is $\text{NPV} = \frac{35}{50}$

☐ E. The F_1-score can be computed as $F_1 = \frac{80}{105}$

☐ F. Since the dataset is balanced (50 positives, 50 negatives), ACC must equal TPR

Solution 3.1

B, C, E

Question 3.2 1 P.

To communicate with the security staff, you must clarify the meaning of different metrics. Which of the following statements about metric relationships are true?

☐ A. The F-measure is the geometric mean of Recall and Precision.

☐ B. The false positive rate (FPR) is the proportion of misclassified samples among all bags flagged by the system.

☐ C. The true positive rate (TPR) is numerically equal to Recall and Sensitivity.

☐ D. If the scanner flags every single bag as a threat, then its Specificity equals 1.

☐ E. The denominator of the negative predictive value (NPV) contains true negatives (TN) and false negatives (FN).

☐ F. Specificity equals $1 - \text{FPR}$.

Solution 3.2
C, E, F

Question 3.3 2 P.
The scanner outputs a continuous confidence score $s(\boldsymbol{x})$. You consider lowering the threshold θ to make the system more sensitive to potential threats. Which statements are correct?

☐ A. The set of bags predicted as positive under a lower threshold is a superset of those predicted as positive under the original threshold.

☐ B. The true positive rate can only increase or stay the same when lowering the threshold.

☐ C. The false positive rate can only decrease or stay the same when lowering the threshold.

☐ D. The specificity can only decrease or stay the same when lowering the threshold.

☐ E. The ROC operating point moves up and to the left when lowering the threshold.

☐ F. Lowering the threshold must increase accuracy, regardless of the class distribution in the luggage.

Solution 3.3
A, B, D

Question 3.4 2 P.
You analyze the Receiver Operating Characteristic (ROC) behavior of the scanning software. Which statements are correct?

☐ A. For a fixed threshold, the scanner corresponds to one point in the ROC plane, while varying θ yields an ROC curve.

☐ B. The diagonal line from $(0, 0)$ to $(1, 1)$ represents random guessing and has $\text{AUC} = 0.5$.

☐ C. AUC is the area under the ROC curve and summarizes performance over all possible threshold settings.

☐ D. If the proportion of bags containing prohibited items changes, the ROC curve must change.

☐ E. Since the ROC curve is independent of the priors, it is also independent of the chosen threshold.

☐ F. The point $\text{FPR} = 0.5,\ \text{TPR} = 0.5$ implies $\text{ACC} = 0.5$ for any class priors.

Solution 3.4
A, B, C, F

Question 3.5 2 P.
The scanner now classifies items into K specific categories (e.g., Electronics, Liquids, Sharp Objects). You examine the K-class confusion matrix $[n_{ij}]$. Which statements are correct?

☐ A. For class Ω_κ, recall_κ uses the row sum (total true instances of that item) in its denominator.

☐ B. For class Ω_κ, precision_κ uses the row sum in its denominator.

☐ C. UAR is a weighted average of per-class recalls, where each class is weighted by its sample count.

☐ D. If every entry in the matrix n_{ij} is doubled, the Recognition Rate (RR) changes because N has changed.

☐ E. If you swap the roles of reference and hypothesis, the Recognition Rate stays unchanged.

☐ F. If you swap the roles of reference and hypothesis, the roles of recall_κ and precision_κ are exchanged.

Solution 3.5
A, E, F

4 Feature Transforms

You are a Lead Data Scientist at a medical research facility developing an automated system to classify K **different tissue types** from high-dimensional biopsy scans. The raw data consists of thousands of features, many of which are redundant or highly correlated. To ensure the diagnostic tool is both accurate and computationally efficient, you must design a pipeline using Linear Discriminant Analysis (LDA) and Principal Component Analysis (PCA). (8 P.)

Question 4.1 1 P.

To begin your pipeline, you consider using rank-reduced LDA for dimensionality reduction. Which of the following statements are correct regarding this method?

- ☐ A. The method selects projection directions that maximize separation between class centroids.
- ☐ B. The PCA step primarily reduces within-class variance by removing low-variance directions inside each class.
- ☐ C. The method constructs a lower-dimensional discriminative subspace that preserves directions most relevant for class separability.
- ☐ D. Rank-Reduced LDA can achieve higher classification accuracy than PCA even when PCA retains more variance.
- ☐ E. The method selects directions that maximize overall variance.

Solution 4.1

A, C, D

Question 4.2 1 P.

Your colleague suggests using standard PCA instead of LDA to differentiate the tissue types. Which of the following statements about PCA and LDA are correct in this context?

- ☐ A. PCA is supervised and thus naturally identifies tissue categories.
- ☐ B. LDA uses class labels (tissue types) to guide the transformation.
- ☐ C. PCA maximizes class separability by default.
- ☐ D. LDA maximizes between-class variance relative to within-class variance.
- ☐ E. PCA guarantees optimal classification for your diagnostic tool.

Solution 4.2

B, D

Question 4.3 2 P.

To handle the varying scales of different biopsy markers, you apply the LDA sphering transform:

$$\phi(x) = D^{-1/2} U^T x$$

This step is designed primarily to achieve which of the following?

☐ A. Transform class-conditional covariance matrices so that all classes share the identity covariance matrix.

☐ B. Remove correlations among features to simplify distance-based classification.

☐ C. Maximize between-class scatter relative to within-class scatter.

☐ D. Guarantee dimensionality reduction without losing discriminative information.

☐ E. Ensure class means become orthogonal in transformed space.

Solution 4.3

A, B

Question 4.4 2 P.

When choosing a distance metric to determine the final tissue diagnosis, which of the following geometric interpretations are correct?

☐ A. The Mahalanobis distance can be interpreted as measuring Euclidean distance in a transformed space where feature correlations and unequal variances have been normalized.

☐ B. Projecting data onto directions of maximum variance (PCA) always improves classification, because higher-variance directions contain more discriminative information.

☐ C. If all class-conditional covariance matrices and class prior probabilities are identical, the Bayes-optimal Gaussian classifier reduces to a nearest-mean decision rule under the Mahalanobis distance.

☐ D. If all class-conditional covariance matrices and class prior probabilities are identical, the Bayes-optimal Gaussian classifier reduces to a nearest-mean decision rule under the Euclidean distance.

☐ E. High-variance directions selected by PCA can capture within-class variation instead of class separation, which may lead to greater overlap between classes after projection.

Solution 4.4

A, C, E

Question 4.5 2 P.

Finally, you evaluate the full LDA model with a shared covariance Σ and the sphered transform $\phi(x)$. Which of the following technical details are correct for your final implementation?

☐ A. After applying ϕ, all classes share the identity covariance matrix. In this space, the LDA decision rule with equal priors is equivalent to nearest-centroid classification using L1 distance.

☐ B. The sphering transform ϕ inherently performs rank reduction from $\mathbb{R}^d$ to $\mathbb{R}^{K-1}$ because the between-class scatter matrix has rank at most $K-1$.

☐ C. In rank-reduced LDA, the projection matrix Φ is obtained by solving a generalized eigenvalue problem and the resulting discriminant subspace has dimension at most $K-1$, regardless of the original feature dimension d.

☐ D. After sphering the data, the LDA decision rule involves squared Euclidean distances and therefore yields quadratic decision boundaries in the transformed feature space.

☐ E. The Fisher Transform viewpoint shows that maximizing the Rayleigh quotient $J(r) = \frac{r^\top S_B r}{r^\top S_W r}$ leads to the same linear boundaries as LDA. For the two-class case, the optimal direction satisfies $r^* \propto S_W^{-1}(\mu_1 - \mu_2)$.

Solution 4.5

C, E

5 Optimization

Question 5.1 6 P.

We consider the following convex optimization problem where we seek to find the point on a line closest to the origin (minimizing energy):

$$\begin{aligned} \text{minimize} \quad & \tfrac{1}{2}\|x\|_2^2 = \tfrac{1}{2}(x_1^2 + x_2^2) \\ \text{subject to:} \quad & x_1 + x_2 \geq 4 \end{aligned}$$

To solve this via duality, we systematically derive the **Lagrange dual function** $g(\lambda) = \inf_x L(x, \lambda)$.

1. **Step 1: Standard Form.**
   Optimization literature typically defines the Lagrangian for inequality constraints in the form $f_i(x) \leq 0$. Which inequality represents the constraint $f_1(x) = x_1 + x_2 \geq 4$ in this standard form? (1 P.)
   - ☐ A. $x_1 + x_2 - 4 \leq 0$
   - ☐ B. $4 - x_1 - x_2 \leq 0$
   - ☐ C. $x_1 + x_2 - 4 = 0$
   - ☐ D. $-(x_1 + x_2) \leq -4$
   - ☐ E. $x_1 \geq 4 - x_2$

2. **Step 2: The Lagrangian.**
   Based on the standard form $f_1(x) \leq 0$ identified above, what is the correct Lagrangian function $L(x, \lambda)$? (1 P.)
   - ☐ A. $L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) - \lambda(x_1 + x_2 - 4)$
   - ☐ B. $L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \lambda(4 - x_1 - x_2)$
   - ☐ C. $L(x, \lambda) = (x_1^2 + x_2^2) + \lambda(4 - x_1 - x_2)$
   - ☐ D. $L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \lambda(x_1 + x_2 - 4)^2$
   - ☐ E. $L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \sum_{i=1}^{2} \lambda_i(4 - x_i)$

3. **Step 3: Minimization w.r.t Primal Variables.**
   To calculate the dual function $g(\lambda) = \inf_x L(x, \lambda)$, we minimize L with respect to x by setting the gradient $\nabla_x L$ to zero. What is the resulting relationship between x and λ? (2 P.)
   - ☐ A. $x_1 = 0$, $x_2 = 0$
   - ☐ B. $x_1 = -\lambda$, $x_2 = -\lambda$
   - ☐ C. $x_1 = \lambda$, $x_2 = \lambda$
   - ☐ D. $x_1 = 4 - x_2$
   - ☐ E. $\nabla_x L = 2x - \lambda = 0 \implies x = \lambda/2$

4. **Step 4: The Dual Function.**
   By substituting the optimal x expressions back into the Lagrangian, we obtain the dual function $g(\lambda)$. Which of the following is the correct expression? (2 P.)
   - ☐ A. $g(\lambda) = \lambda^2 - 4\lambda$
   - ☐ B. $g(\lambda) = 4\lambda - \lambda^2$

☐ C. $g(\lambda) = 4\lambda - 2\lambda^2$

☐ D. $g(\lambda) = 2\lambda^2 + 4$

☐ E. $g(\lambda) = -\frac{1}{2}\lambda^2 + 4\lambda$

Solution 5.1

1. **B**: The standard form requires $f_i(x) \leq 0$. Rearranging $x_1 + x_2 \geq 4$ yields $4 - x_1 - x_2 \leq 0$.
2. **B**: The Lagrangian is defined as $L(x, \lambda) = f_0(x) + \sum \lambda_i f_i(x)$. Using the result from Step 1, this is $\frac{1}{2}\|x\|^2 + \lambda(4 - x_1 - x_2)$.
3. **C**: Taking the partial derivatives: $\frac{\partial L}{\partial x_1} = x_1 - \lambda = 0 \Rightarrow x_1 = \lambda$. Similarly, $x_2 = \lambda$.
4. **B**: Substituting $x_1 = \lambda, x_2 = \lambda$ back into L:

$$g(\lambda) = \frac{1}{2}(\lambda^2 + \lambda^2) + \lambda(4 - \lambda - \lambda) = \lambda^2 + 4\lambda - 2\lambda^2 = 4\lambda - \lambda^2$$

This process is analogous to the derivation of the SVM dual.

6 Expectation Maximization

You are a geophysicist analyzing a large dataset of seismic vibrations. Because the different geological sources of these vibrations are not explicitly labeled, you utilize a **Gaussian Mixture Model (GMM)** to perform density estimation and discover latent patterns in the activity. You apply the **Expectation-Maximization (EM)** algorithm to refine your model parameters. (8 P.)

Question 6.1 1 P.

As you structure the latent variables for your seismic model, you evaluate how the model handles unobserved assignments. **Which of the following statements are *incorrect*?** (Select all that apply.)

☐ A. The Observed-data log likelihood is guaranteed to increase whenever the responsibilities are updated.

☐ B. If the latent variables Z (the specific seismic source) were observed, maximum likelihood estimation would reduce to fitting independent Gaussian distributions to disjoint subsets of the data.

☐ C. The Observed-data log likelihood is Convex in the mixture parameters, since it is obtained by marginalizing over the latent variables.

☐ D. Knowing the values of the latent variables removes the need for the EM algorithm, because the resulting maximum likelihood problem admits a closed-form solution.

☐ E. When component assignments are treated as known, the log-likelihood separates into independent terms for each component, whereas when assignments are unknown, the log-likelihood involves a log of a sum over components.

Solution 6.1

A, C

Question 6.2 1 P.

To quantify the probability that a specific tremor belongs to "Volcanic Activity" versus "Tectonic Shift," you calculate the responsibilities $\gamma(z_{nk})$. Which of the following statements are correct? (Select all that apply.)

☐ A. Responsibilities are posterior probabilities of latent variables.

☐ B. Responsibilities sum to one over mixture components for each data point.

☐ C. Responsibilities correspond to hard cluster assignments (each tremor is assigned to exactly one source).

☐ D. Responsibilities depend on current parameter estimates (π_k, μ_k, Σ_k).

☐ E. Responsibilities are optimized in the M-step of the algorithm.

Solution 6.2

A, B, D

Question 6.3 2 P.

While training your seismic model, you monitor the convergence of the EM algorithm. Which of the following statements are true about the algorithm's behavior? (Select all that apply.)

- ☐ A. The choice of initial values for the seismic component parameters (e.g., initial mean vibration frequency) affects the final estimates.
- ☐ B. The EM algorithm is guaranteed to converge.
- ☐ C. The EM algorithm is guaranteed to converge to the global maximum of the likelihood.
- ☐ D. The parameter estimates obtained at convergence are guaranteed to be the global Maximum Likelihood Estimates.

Solution 6.3

A, B

Question 6.4 2 P.

You consider incorporating historical geological data as a prior distribution $p(\theta)$ using Maximum A Posteriori (MAP) estimation. Which of the following statements regarding MLE and MAP are correct?

- ☐ A. The MLE depends only on the likelihood function, whereas the MAP estimator additionally depends on the prior distribution $p(\theta)$.
- ☐ B. If your prior for seismic activity is uniform, the MAP estimator becomes identical to the MLE.
- ☐ C. MAP estimation can be formulated as maximizing the sum of the log-likelihood and the log-prior.
- ☐ D. Because the MAP estimator utilizes prior geological knowledge, it is mathematically guaranteed to always provide a better estimate of the true parameters than MLE.
- ☐ E. For a massive dataset of seismic records, the influence of the prior on the MAP estimator necessarily disappears, regardless of the choice of prior.

Solution 6.4

A, B, C

Question 6.5 2 P.

Your model occasionally produces errors where a component "collapses." **Which of the following technical statements are correct** regarding the refinement of your GMM? Select all that apply.

- ☐ A. The EM algorithm directly optimizes seismic classification accuracy during the training process.
- ☐ B. During the EM algorithm, the expected complete-data log-likelihood is maximized with respect to the model parameters while keeping the responsibilities fixed.
- ☐ C. A fixed point of the EM algorithm may correspond to a saddle point of the observed-data log-likelihood rather than a strict local maximum.
- ☐ D. The EM algorithm can converge to degenerate solutions in which a component collapses onto a single seismic event with a vanishing covariance matrix.

☐ E. A GMM trained via EM performs density estimation; identifying a specific tremor's source requires an additional decision rule based on posterior probabilities.

Solution 6.5

B, C, D, E